

Element A

Problem Identification

Our problem is that the komodo dragons at the Nashville zoo are not moving around their exhibit enough at the zoo to achieve optimal health. The problem is they will sit in one part of the exhibit which fails to emulate a natural life where the komodo dragons would sprint to get to food or evade predators. This is important, because this sedentary lifestyle is a problem as it is harmful to the komodo dragon's physical and mental health, and the dragons health is important as it is integral to their quality of life which is part of Nashville Zoo's mission of conservation and to the enjoyment of patrons.

In order to fix this issue the zoo needs a design that will have the komodo dragons sprinting every couple of weeks for a short amount of time as this will emulate the komodo dragon's natural behavior as stated by Smithsonian's National Zoo and Conservation Biology Institute of waiting to ambush prey and then sprinting out at them and their behavior of only eating about once a month as shown by the Jacksonville zoo.

The komodo dragons cant sprint up to 13 mph and walk at 3 mph as evidenced by the San Diego zoo Wildlife Library, so having them get up to these speeds even once a month would be much more optimal than their current habits of sitting and waiting for food to be thrown to them. The zookeepers said they would like for the komodo dragons to run across the habitat 2 - 4 times.

Any solutions need to be easily cleaned as the zoo keepers are worried about any bacteria or mold growing in the habitat.

The komodo dragons are sometimes fed by tongs by the zookeepers but this puts the zookeepers at risk of attack by the dragons and the dragons fail to meet the natural speeds they would get to when ambushing.



(above, komodo dragon exhibit at the Nashville Zoo, photo from brief)

Stakeholders

Primary:

- **Komodo Dragons** will be the ones using this device and most affected by the success of our solution. Their mental and physical health is tied to this product.
- **Zoo keepers** will be the ones controlling the device and their safety is also tied to the success of the solution as they must be able to control the product out of reach of the komodo dragons due to the ferocity, jaw strength and venom of the komodo dragons.
- **Zoo Managers** - The managers are responsible for managing and planning all zoo operations and because of this the success of this project is paramount to their job's success.
- **Komodo Dragon Habitat** - The project will be set up within the komodo dragon habitat and the habitat must remain undamaged.
- **E4USA Program** - E4USA is a primary stakeholder as many of the resources are supplied by E4USA and the reputation of E4USA is based on the success of this project.

Secondary:

Patrons - The patrons are impacted as they come to the zoo to see the komodo dragons and would like to see healthy dragons. Our project would better the health of the komodos which would meet the patrons' expectations and they would come to see the komodo dragons more often.

In our interview with the primary stakeholders, the zookeepers at the Nashville Zoo, who would make use of this solution we asked the following questions. Note not all questions were answered.

2/23/26-2/25/26

- Should the solution be reusable? Should be made so the dragon doesn't really eat it
- How and where would you like it to be placed in the enclosure?
- How much can this solution weigh (without food)?
- Minimum amount of food that should be used if we use food?
- What materials can we use and where is the best place to find/get/buy them?
- Allergies?
- What scares it?
- What do yall feed it? Dead goat and rabbits (40lbs)
- What stimulates komodo dragons most food, scent, or movement? Scent is the primary stimulus for Komodo dragons
- Do they rely more on sight or or smell
- Do they prefer being in shade, sun, rocks, water or open space
- How far away can they detect food with sight or smell? Komodo dragons can detect food primarily through an exceptional sense of smell, allowing them to track carrion from up to 2.5 to 6 miles (4 to 9.5 kilometers) away.
- What are the animals' active hours?
- Preferred size of the solution?
- Would you like the device to be placed from above?

- Should this be something the zoo keeper can control or should the keeper just place it in the exhibit and let the Komodo Dragon use it? **Either but keeper would be on stand by**
- Any Previous enrichment toys?
- What irritates the komodo?
- What is the komodo dragon's Weight?
- What is the komodo dragon's age?
- What are their Genders?
- How fast is the komodo dragon? **capable of reaching speeds up to 13 mph in short**
- Should it be usable by multiple of them at the same time or one at a time? **Multiple**
- Does the dragon have a preference for season?
- Do they currently have any enrichment toys in the exhibit? Whether working or broken
- Habitat size
- Do the dragons compete over enrichment or use it one at a time? **One at a time as they are territorial**
- What is their average feeding schedule? **Feed Lil Sebastian once a month.**
- Are there seasonal behavior changes? **They just finished a fast Nov-Feb.**
- Do you want this to encourage climbing, pulling, tracking, or swimming?

2/25/26 Shared Questions with the other group doing this project:

- Does the toy have to stay standing after they are done, or can it be a consumable and then remade from a mold?

- If it's a mold, what is the preferred material? What materials are available to fill the mold?
- What is considered pursuit behaviors?
- Should the enrichment toy be warm or cold? / Does the temperature matter?
- Will the enrichment toy be used when the komodo dragons are not outside?
- Does it need to be water proof or water resistant? Does not need to be waterproof but can go in water
- How many calories do the komodo dragons consume per day?
 - What is the maximum amount of food that you would want to put in an enrichment toy?
- How many enrichment toys do we need to manufacture? One
- What is the ideal mobility of the toy?
- How long does the toy need to last?
- What is the bite force of the strongest komodo dragon here? The bite force is 1150N
- What is considered a treat for the komodo dragons and what is considered their normal diet?
- Are there any materials or items that the komodo dragons don't consume/could harm the komodo Dragon? This was researched by us but Zinc plating is poisonous so anything Zinc Plated will be far from the dragons.
- What scents do the Komodo dragons like? And dislike?

Problem Importance: Our problem is important because the komodo dragons need to be moving around to keep up their health as part of the Zoo's goal is to promote the welfare of the dragons.

Additionally, the Dragons need to be running as it is part of their natural lifestyle and can help the dragons to live up to 30.